NOVEMBER 2026

Private Equity Essentials: Terminology, Structure and Concepts

A reference map of private-market vocabulary, organized from fund structure through economics, metrics, deals, capital structure, terms, exits and **regulation**. Cross-references to the Leviathan model are marked ▸.

10
domains in the reference map

3
worked examples: waterfall, LBO, EV bridge[2]

$1,406M
Leviathan holdco used as the running example[1]

00 The map

01
Structure
GP, LP, ManCo, fund vehicle; feeders, parallels, AIVs, blockers; SPVs and co-invest; continuation vehicles; holdco and sleeves

02
Lifecycle
Fundraise and closes; investment period; harvest; capital calls and distributions

03
Economics
Management fee; carry, hurdle and catch-up; European vs American waterfall; clawback

04
Metrics
IRR and MOIC; TVPI, DPI, RVPI; public market equivalent; the J-curve

05
Strategies
VC, growth, buyout; carve-outs and take-privates; secondaries and co-invest; credit and distressed

06
Deal process
Teaser, CIM, IOI, LOI; diligence and QoE; SPA, signing and closing

07
Valuation
Enterprise-to-equity bridge; multiples, DCF and LBO; pre-and post-money

08
Capital structure
Revolver, TLB, unitranche; second lien, mezzanine, PIK; preferred and common

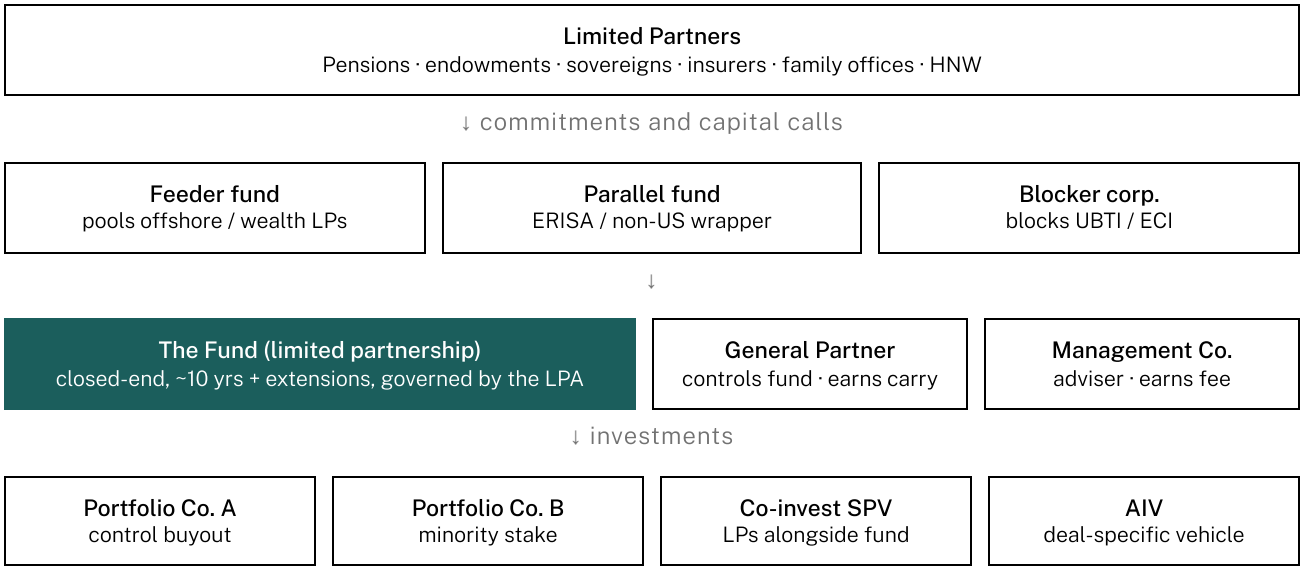
09
Equity terms
Liquidation preference; anti-dilution; drag, tag and ROFR; board and information rights

10
Exits & regulation
Strategic, sponsor, IPO, continuation; CFIUS, ITAR, FOCI; Advisers Act, ERISA; defense contracting

01 Players and Fund Structure[1]

How the entities connect

Capital flows down, proceeds flow back up through the waterfall



Term	Definition	Why it matters
GP (General Partner)	Entity that controls the fund and makes investment decisions; has unlimited liability (usually through an LLC)	Earns carry; legally responsible
LP (Limited Partner)	Investor in the fund (pensions, endowments, sovereigns, insurers, family offices, HNW); liability limited to its commitment	Supplies roughly 95–99% of capital
Management Company (ManCo)	Adviser entity that employs the team and receives management fees	Registered investment adviser; holds the economics of running the firm
Fund (Limited Partnership)	Usually a Delaware or Cayman LP; closed-end, about 10 years plus extensions	The legal pool of capital
LPA (Limited Partnership Agreement)	The governing contract: economics, governance, term, key person provisions	Everything is negotiated here
PPM (Private Placement Memorandum)	Offering document describing strategy, team, risks and terms	Disclosure document
Subscription documents	LP's binding commitment plus investor qualification forms	Create the legal commitment
Side letter	Bilateral LP-specific terms (fee breaks, reporting, co-invest rights, ESG or sanctions exclusions)	Usually come with an **MFN** clause
MFN (Most Favored Nation)	LP can elect any better term given to a similar-size LP	Limits preferential deals
LPAC (LP Advisory Committee)	Committee of large LPs that approves conflicts, valuations and extensions	Governance check on the GP

Term	Definition	Why it matters
Feeder fund	Vehicle that pools investors (e.g. offshore, or wealth platforms) into the main fund	Tax and regulatory access
Parallel fund	Separate vehicle investing side-by-side pro rata (e.g. for ERISA, non-US or tax-sensitive LPs)	Same deals, different wrapper
AIV (Alternative Investment Vehicle)	Deal-specific vehicle used for tax or regulatory reasons	Routes particular investments
Blocker corporation	Corporation interposed to block UBTI (US tax-exempts) or ECI (non-US investors)	Tax efficiency for specific LPs
SPV (Special Purpose Vehicle)	Single-asset vehicle, often for co-investment or one deal	Concentrated exposure, deal-by-deal economics
Co-investment	LP invests directly alongside the fund in one deal, usually at low or zero fee and carry	Lowers blended fees; lets the GP write bigger checks
Fund of funds	Fund that invests in other PE funds	Diversification at the cost of two layers of fees
SMA (Separately Managed Account)	Single-LP custom mandate	Custom terms, typically large LPs
Evergreen / semi-liquid fund	Open-ended vehicle with periodic subscriptions and redemptions (common in wealth channels)	No fixed term; liquidity gates
Continuation vehicle (CV)	GP-led secondary: assets move from an old fund to a new vehicle, and existing LPs choose to roll or sell	Extends the hold on winners
Holdco	Holding company owning stakes across sub-portfolios or sleeves	▸ Leviathan = $1,406M holdco across three sleeves
Sleeve	Distinct sub-strategy or sub-fund within a vehicle	▸ IQT-mirror / Weapons / Consumer
GP commitment	GP's own capital in the fund (typically 1–5%)	Alignment ("skin in the game")
Key person clause	Investment period suspends if named individuals leave	Protects LPs against team risk
No-fault divorce	LPs can remove the GP or end the investment period by supermajority without cause	Ultimate LP lever

02 Fund Lifecycle

Fundraise
Year 0 · 12–18 months
First close, final close, LPA and side letters

Investment period
Years 1–5
Capital calls; new platforms; follow-on reserve set

Value creation
Years 3–8
Add-ons, follow-ons, operating improvements

Harvest
Years 4–10
Exits and distributions; DPI builds

Extension
Years 10–12
Tail-end sales or continuation vehicle; wind-down

Term	Definition
Commitment	Total capital an LP agrees to provide over the fund's life
First / final close	First and last dates LPs can join; later closers pay an equalization interest
Capital call (drawdown)	GP requests a portion of commitments, usually with about 10 business days' notice
Paid-in capital (PIC)	Cumulative capital actually called
Unfunded commitment	Commitment not yet called
Dry powder	Industry-wide or fund-level uncalled capital available to invest
Investment period	Window (about 5 years) in which new platforms can be acquired
Harvest period	Post-investment period: follow-ons and exits only
Recycling	Reinvesting early proceeds (up to a cap) instead of distributing them
Distribution	Cash or stock returned to LPs
Distribution in kind	Distributing shares (e.g. post-IPO stock) instead of cash
Subscription line (capital call facility)	Bank loan secured by LP commitments; delays calls and **inflates IRR**
NAV loan	Fund-level borrowing secured by portfolio NAV; used for liquidity or follow-ons
Vintage year	Year of first investment or first close; used for peer comparison
Follow-on reserve	Capital set aside for existing portfolio companies ▸ IQT $25M of $250M

03 Fund Economics[2]

3.1 Fees

Term	Definition	Typical
Management fee	Annual fee to the ManCo	1.5–2.0% of **committed** capital during the investment period, stepping down to **invested** capital or a lower rate afterward ▸ model uses 2% of committed for the full life
Fee offset	Share of transaction and monitoring fees credited against the management fee	80–100% offset today
Transaction / monitoring fees	Fees the GP charges portfolio companies	Increasingly offset or banned
Organizational expenses	Fund formation costs charged to the fund (capped)	e.g. $1–3M cap
Fund expenses	Audit, admin, legal, broken-deal costs	Charged to the fund

3.2 Carry and Hurdle

Term	Definition
Carried interest (carry)	GP's share of profits, typically 20% (10–30%)
Preferred return (hurdle)	Minimum LP return before carry is paid, usually 8% compounded annually
Catch-up	After the pref, the GP takes a large share (often 100%) until it has caught up to its 20% of **total** profits
European (whole-fund) waterfall	Carry only after LPs have received **all contributed capital plus pref** across the entire fund. LP-friendly
American (deal-by-deal) waterfall	Carry paid on each realized deal (usually after returning that deal's cost plus allocated expenses). GP-friendly; needs a clawback
Clawback	GP must return carry already received if the final fund result doesn't support it
Escrow / holdback	Part of carry withheld to secure the clawback
Hard vs soft hurdle	Hard: carry only on profits **above** the hurdle (no catch-up). Soft: once the hurdle is met, carry applies to all profit (with catch-up)

3.3 Worked Waterfall: $100M fund, $250M gross proceeds, 5-year single exit

Pref = 100 × (1.08^5 − 1) = **$46.93M**. Total profit = $150M.

Tier	Full catch-up (soft hurdle)	No catch-up (hard hurdle) ▸ Leviathan
1. Return of capital (LP)	100.00	100.00
2. Preferred 8% (LP)	46.93	46.93
3. GP catch-up (100%)	11.73 (= 46.93 × 20/80)	—
4. Remaining split 80/20	LP 73.07 / GP 18.27	LP 82.46 / GP 20.61
GP carry total	**30.00** (= 20% of 150)	**20.61**

Tier	Full catch-up (soft hurdle)	No catch-up (hard hurdle) ▸ Leviathan
LP total (before fees)	220.00	229.39

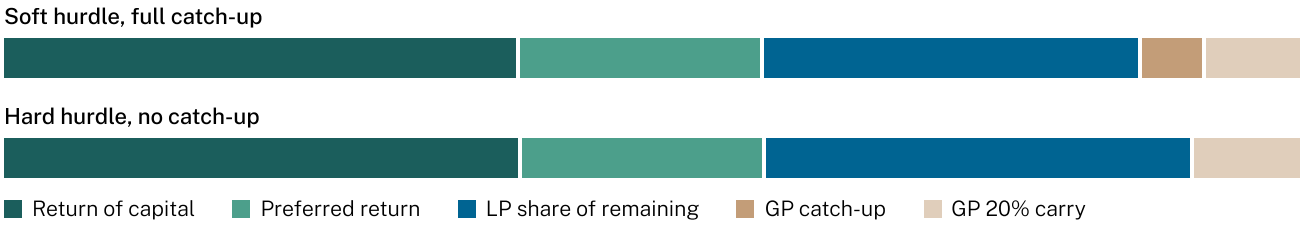
Fees are then deducted: 2% × $100M × 5 years = $10M. The difference between the two waterfalls (about $9.4M here) grows with performance.

$30.0M
GP carry with full catch-up (20% of $150M profit)

$20.6M
GP carry, hard hurdle, no catch-up ▸

$9.4M
value of the catch-up to the GP in this example

Distribution of $250M gross proceeds[2]
$M, before management fees



04 Performance Metrics[1]

Metric	Formula	Notes
MOIC (multiple on invested capital)	Total value ÷ invested capital	Ignores time. Gross (deal level) vs net (to LP)
IRR (internal rate of return)	Discount rate r such that $\Sigma\ CF_t/(1+r)^t = 0$	Time-weighted. Sensitive to timing and to subscription lines. Single-exit shortcut: MOIC^(1/T) − 1
TVPI (total value to paid-in)	(Cumulative distributions + NAV) ÷ PIC	Net multiple, realized plus unrealized
DPI (distributions to paid-in)	Cumulative distributions ÷ PIC	"Cash-on-cash." What LPs care about most in tight markets
RVPI (residual value to paid-in)	NAV ÷ PIC	Unrealized portion; TVPI = DPI + RVPI
PME (public market equivalent)	Compares PE cash flows with an index	Answers "did I beat the index on the same cash-flow timing?"
KS-PME (Kaplan–Schoar)	$\Sigma\ Dist_t \times (I_T/I_t) + NAV_T$, divided by $\Sigma\ Call_t \times (I_T/I_t)$	>1.0 = beat the index. Single in/out: MOIC ÷ (1+r_index)^T ▸ 5-yr Base 2.42 ÷ 1.09^5 = 1.57x vs S&P 500
Direct Alpha	IRR of index-compounded cash flows	Annualized excess return over the index
Long–Nickels PME (LN-PME / ICM)	Implied IRR from investing the same flows in the index	Can break down with large distributions
J-curve	Early negative returns (fees, write-downs) before value creation and exits turn the curve up	Subscription lines flatten it artificially
Gross-to-net spread	Gross IRR − net IRR	Typically 4–7 percentage points for buyout (fees, carry, expenses)
Loss ratio	% of capital invested in deals returning < 1.0x	Measures underwriting discipline
Quartile ranking	Performance versus same-vintage peers	Top quartile matters; dispersion is large in VC

Common traps: IRR can be gamed with subscription lines and quick flips. MOIC ignores time. Interim NAVs are Level 3 marks (judgment). Always look at DPI and PME together.

05 Strategies

Strategy	Profile	Typical control / leverage	Return driver
Venture capital (pre-seed → Series A/B/C → late)	Pre-profit, high growth, power-law outcomes	Minority, no leverage	A few outsized winners
Growth equity	Revenue scaling, near or at profitability	Minority, little leverage	Revenue growth
Buyout / LBO	Mature, cash-generative	Control, 4–7x debt/EBITDA	EBITDA growth, deleveraging, multiple
Take-private (P2P)	Buying a listed company	Control	Re-rating plus operational improvement
Carve-out / spin-off	Buying a division from a corporate parent	Control or minority	Standalone efficiency; TSAs
Buy-and-build / roll-up	Platform plus add-on acquisitions	Control	Multiple arbitrage (buy add-ons at 6x, consolidate at 12x)
Minority / structured equity	Preferred with downside protection	Minority, structured	Contractual returns plus upside ▸ Ellwood, ARA
Pre-IPO / crossover	Last private round before listing	Minority	Discount to IPO price ▸ IAI cornerstone, Syntiant
PIPE	Private investment in public equity	Minority	Discount plus terms ▸ Infleqtion
Secondaries, LP-led	Buying LP fund interests	Passive	Discount to NAV, shorter J-curve
Secondaries, GP-led	Continuation vehicles, strip sales	Varies	Extending winners
Private credit / direct lending	Senior loans to sponsor-backed companies	Lender	Floating-rate spread
Mezzanine	Subordinated debt plus warrants	Lender	12–15% blended
Distressed / special situations	Stressed balance sheets	Loan-to-own	Restructuring
Infrastructure / real assets	Long-duration contracted cash flows	Control	Yield plus inflation linkage

06 Deal Process

1. Sourcing → NDA
Proprietary or banker-led; confidentiality signed

2. Teaser / CIM → IOI
Anonymous summary, full book, non-binding range

3. Mgmt pres. + VDR → LOI
Data room; price, structure, exclusivity

4. Confirmatory diligence
QoE, legal, tax, commercial, regulatory

5. SPA → sign → close
HSR / CFIUS clearance, then funds flow

Term	Definition
Proprietary deal	Sourced outside a banker auction; less competition
Auction (broad / limited)	Banker-run process with rounds of bids
Teaser	Anonymous 1–2 page summary
CIM (Confidential Information Memorandum)	Detailed sell-side book
IOI (Indication of Interest)	Non-binding first-round valuation range
LOI (Letter of Intent) / term sheet	Second-round proposal; usually grants **exclusivity**
Data room (VDR)	Repository of diligence documents
QoE (Quality of Earnings)	Accounting diligence normalizing EBITDA
Commercial diligence	Market, customers, competitive position ▸ contract inventory and program build
Confirmatory diligence	Legal, tax, IT, environmental, insurance, HR and regulatory diligence after the LOI
SPA / APA	Stock or Asset Purchase Agreement
Reps & warranties	Seller statements of fact; breaches are indemnified
R&W insurance (RWI)	Policy covering rep breaches; reduces escrow needs
Indemnification / escrow	Seller's post-close liability and the cash held to secure it
MAC / MAE	Material adverse change clause allowing walk-away
Locked box vs completion accounts	Fixed historical balance sheet vs post-close true-up
Earn-out	Contingent consideration tied to future performance
Rollover equity	Sellers or management reinvest part of their proceeds
TSA (Transition Services Agreement)	Parent keeps providing services to a carve-out temporarily
HSR	US antitrust pre-merger filing
Sign-to-close	Gap between signing and closing, covered by covenants

07 Valuation

Enterprise value to equity value bridge

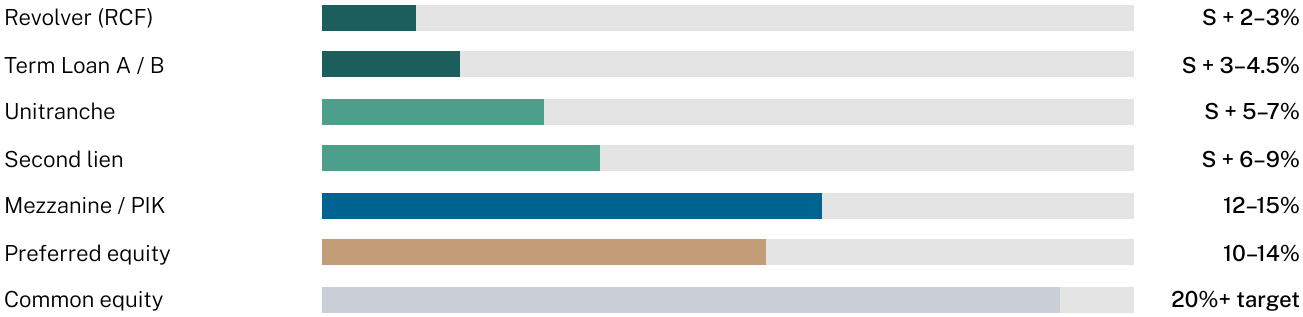
Equity Value = Enterprise Value − Debt − Preferred − Minority interest − Debt-like items (leases, pensions, deferred rev*) + Cash

Term	Definition
EBITDA / Adjusted EBITDA	Operating cash proxy. "Adjusted" removes one-offs, and is where sellers inflate the number
Run-rate / pro forma EBITDA	Annualized latest results plus synergies or acquisitions
EV/EBITDA, EV/EBIT, EV/Revenue	Core multiples ▸ LHX 15.1x and RHM 17.3x EV/EBITDA; Saronic ~46x EV/revenue
Trading comps	Multiples of similar listed companies
Precedent transactions	Multiples paid in comparable M&A (include a control premium)
DCF	PV of free cash flow plus terminal value (Gordon growth or exit multiple)
LBO analysis	Maximum price a sponsor can pay and still hit its target IRR
Pre-money / post-money	Value before / after new money. Post = Pre + new investment. Ownership = Check ÷ Post ▸ Hermeus $35M ÷ $1.0B = 3.5%
Fully diluted	Counts options, warrants and convertibles
Option pool shuffle	Pool created in the pre-money, so the dilution falls on existing holders
Down round / flat round	New price below / equal to the prior round
409A	Independent common-stock valuation used for option strikes
ASC 820 / Level 3	Fair-value accounting. Private marks are model-based (Level 3)
Probability-weighted value	$\Sigma\, p_i \times \text{Value}_i$ across scenarios ▸ holdco 7-yr Expected 4.63x = 45% × 3.56x + 25% × 7.79x + 15% × 1.05x + 15% × 6.14x
Control premium / minority discount / DLOM	Adjustments for control and lack of marketability

08 Capital Structure and LBO Mechanics[2]

The capital stack: senior to junior[3]

Indicative cost of capital; bar = midpoint. S = SOFR spread for floating-rate debt.



Senior claims are paid first and cost least; common equity is paid last and targets the highest return. Pricing varies with rates and credit quality.

Stack (senior → junior): Revolver → Term Loan A → Term Loan B / Unitranche → Second lien → Senior/sub notes → Mezzanine / PIK → HoldCo PIK → Preferred equity → Common equity (sponsor + management).

Term	Definition
Revolver (RCF)	Drawable working-capital line
TLA / TLB	Amortizing bank loan / institutional, mostly bullet, floating rate
Unitranche	Single blended senior loan from a direct lender
Second lien	Junior secured claim on the same collateral
Mezzanine	Subordinated debt with warrants or equity kicker
PIK (payment-in-kind)	Interest accrues to principal instead of being paid in cash
Leverage	Debt ÷ EBITDA (total, senior, net)
Interest coverage / FCCR	EBITDA ÷ interest; (EBITDA − capex − tax) ÷ fixed charges
Maintenance vs incurrence covenants	Tested quarterly vs only when taking an action
Cov-lite	No maintenance covenants (common in TLB)
Sources & uses	Table showing where the money comes from and where it goes
Dividend recap	Borrow to pay the sponsor a dividend (DPI without an exit)
Deleveraging	Using FCF to repay debt, which increases equity value
Management Incentive Plan (MIP) / sweet equity	Management equity, often 5–15% of common, with ratchets

Worked LBO. Entry EBITDA $50M × 10x = EV $500M, funded with 5x debt ($250M) and $250M equity. Year 5: EBITDA $80M × 10x = EV $800M; net debt paid down to $120M, so equity = **$680M**. · MOIC = 680 ÷ 250 = **2.72x**; IRR = 2.72^(1/5) − 1 = **22.2%** · **Value-creation bridge:** EBITDA growth (30 × 10x) = $300M · Multiple expansion = $0 · Deleveraging = $130M · Total = **$430M** gain

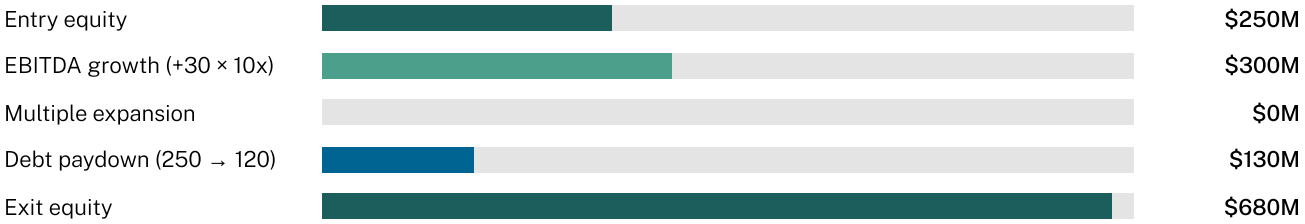
$680M
exit equity value (80 × 10x − 120 net debt)

2.72x
MOIC on $250M sponsor equity

22.2%
IRR over a 5-year hold

LBO value-creation bridge[2]

$M; entry 10x $50M EBITDA with 5x debt



09 Equity Terms (Venture, Growth, Minority)

The terms that protect a minority investor matter most for stakes like the Leviathan ones.

Term	Definition	Investor-friendly version
Liquidation preference	Paid before common in an exit	1x non-participating is standard; >1x or participating is aggressive
Participating preferred	Gets its preference back **and then** shares pro rata	"Double dip"; often capped (e.g. 3x)
Seniority	Order among preferred series	Senior vs pari passu
Anti-dilution	Price protection in down rounds	Full ratchet (strong) vs broad-based weighted average (standard)
Pro rata / preemptive right	Right to keep ownership % in future rounds	Key for follow-on reserves
ROFR	Right of first refusal on share transfers	Controls the cap table
Tag-along (co-sale)	Minority can sell alongside the majority on the same terms	Essential minority protection
Drag-along	Majority can force the minority to sell	Minority wants a price floor or threshold
Protective provisions / vetoes	Consent needed for new senior securities, sale, debt, budget	Key minority control
Board seat / observer	Governance or visibility	▸ Seek one at >15% ownership
Information rights	Monthly or quarterly financials, budget, audit	Basic
Redemption right	Investor can force a buyback after N years	Liquidity path in family or ESOP-owned companies
Put / call options	Contractual exit rights	Used in structured minority deals
Conversion	Preferred converts to common (automatically at a qualified IPO)	Automatic conversion on a qualified IPO is the standard pre-IPO exit path
IPO ratchet	Extra shares if the IPO prices below a threshold	Pre-IPO protection
Warrants	Right to buy shares at a strike	Equity kicker; common in mezzanine and SPAC deals
Convertible note / SAFE	Early instruments converting at the next round, with cap and discount	Seed stage
Pay-to-play	Investors who don't follow on lose preferences	Penalizes free riders
Dividend (cumulative / non-cumulative)	Preferred coupon	Structured preferred usually 8–12% cumulative
Vesting / cliff	Founder and employee equity earned over time	Retention

Dilution and retention: if a fund owns 10% today and new rounds sell 15%, 10% and 10%, then final ownership = 10% × 0.85 × 0.90 × 0.90 = 6.9% (retention 0.69). ▸ The model's 0.68 retention at 10 years for the IQT sleeve encodes the same idea.

10 Exits

Route	Mechanics	Notes
Strategic sale	Sale to a corporate buyer	Synergies mean the highest price. CFIUS limits who can buy
Secondary buyout (SBO)	Sale to another sponsor	Common; "pass the parcel" critique
IPO	Listing; sponsor sells over time	**Lock-up** (usually 180 days), greenshoe, cornerstone investors
Direct listing / SPAC	Alternative ways to list	SPACs are mostly discredited post-2021
Dividend recap	Partial liquidity through debt	No change of control
Continuation vehicle	Sell to a GP-led vehicle	Liquidity for LPs, longer hold for winners
Strip sale / minority sale	Sell part of the stake	Partial DPI
Write-off	Zero	Counts in the loss ratio

11 Value Creation

- **100-day plan**: early priorities after close covering management, KPIs, quick wins and reporting.
- **Levers**: revenue (pricing, new markets, salesforce), margin (procurement, footprint, automation), capital efficiency (working capital, capex discipline), M&A (add-ons), capital structure (refinancing, recaps), multiple (scale, quality and growth re-rating).
- **Operating partners**: in-house operators supporting portfolio companies.
- **Buy-and-build math**: buy a $10M EBITDA platform at 10x, add $10M EBITDA of add-ons at 6x, and the combined $20M exits at 11x = $220M EV versus $160M cost before any organic growth.
- **For defense targets specifically**: capacity expansion ahead of multiyear awards, qualifying new programs, moving from tier-2 to tier-1 status, gaining FMS and export approvals, CMMC compliance, and facility clearances.

12 Regulatory and Legal[5]

12.1 Fund and Investor Regulation (US)

Term	Definition
Investment Advisers Act / RIA / Form ADV	Adviser registration and disclosure
Accredited investor / Qualified purchaser (QP)	Investor wealth thresholds ($1M net worth or $200K income / $5M investments)
3(c)(1) / 3(c)(7)	Investment Company Act exemptions (≤100 beneficial owners / QPs only)
ERISA 25% test / VCOC	Avoid plan-asset status if benefit-plan investors are below 25%
UBTI / ECI	Tax issues for tax-exempt and non-US LPs; solved with blockers
ILPA Principles	LP best-practice standards for terms and reporting
Carried interest taxation	Long-term capital gains treatment after a 3-year holding period (IRC §1061)
QSBS (§1202)	Capital-gains exclusion on qualifying C-corp stock held 5+ years

12.2 National Security and Defense Investing

Term	Definition	Relevance
CFIUS	Committee reviewing foreign investment in US businesses; mandatory filings for critical technology	Determines who can invest and who can buy at exit
TID business	Critical Technology, Infrastructure or sensitive personal Data, which triggers heightened CFIUS review	Most defense and semis targets
ITAR / EAR	Export controls on defense articles (USML) / dual-use items (CCL)	Limits foreign ownership and access
FOCI	Foreign Ownership, Control or Influence. Mitigated via a Board Resolution, SCA, SSA or Proxy Agreement	Required for cleared contractors with foreign investors
FCL (facility clearance)	Clearance needed to perform classified work	Valuable and slow to obtain, which makes it a moat
NDAA §889 / §848 / Blue UAS	Bans on Chinese components; approved drone list	Provenance diligence ▸ DefendEye
CMMC / DFARS 252.204-7012	Cybersecurity requirements for DoD suppliers	Compliance cost and gate
Outbound investment rules	US restrictions on investing in Chinese semis, AI and quantum	Affects allied structures

12.3 Defense Contracting Vocabulary

Term	Definition
PPBE	Planning, Programming, Budgeting and Execution: DoD's budget cycle
Request vs authorization vs appropriation	President's budget ask → NDAA policy limit → actual money. Only appropriations fund contracts
CR (Continuing Resolution)	Stopgap funding at prior-year levels; blocks new starts and slows ramps

Term	Definition
Reconciliation	Budget process used for large one-off funding (e.g. much of Golden Dome)
IDIQ	Indefinite Delivery / Indefinite Quantity. A ceiling contract; revenue comes only from task orders
MYP (Multiyear Procurement)	Congress-approved multiyear buys that give suppliers visibility ▸ Tomahawk, PAC-3
LRIP / FRP	Low-Rate Initial Production / Full-Rate Production
FFP vs cost-plus	Firm Fixed Price (contractor bears cost risk; margin upside) vs Cost-Plus (cost reimbursed plus a fee; lower risk and margin)
FMS vs DCS	Foreign Military Sales (government-to-government through DSCA) vs Direct Commercial Sales
Prime / tier-1 / tier-2 / tier-3	Position in the supply chain ▸ Ellwood, Superior & Grand Valley and Haigh-Farr are tier-2 or tier-3 chokepoints
Sole source / J&A	Non-competitive award with justification. Strong moat
Recompete	Re-bidding when a contract expires: a revenue cliff risk
OTA (Other Transaction Authority)	Fast prototyping contracts outside the FAR (DIU, consortia)
SBIR / STTR Phase I / II / III	Small business R&D awards. Phase III = production without re-competition
Black / classified budget	Unacknowledged programs; the IC funds NIP and MIP ▸ IQT funded within the NIP

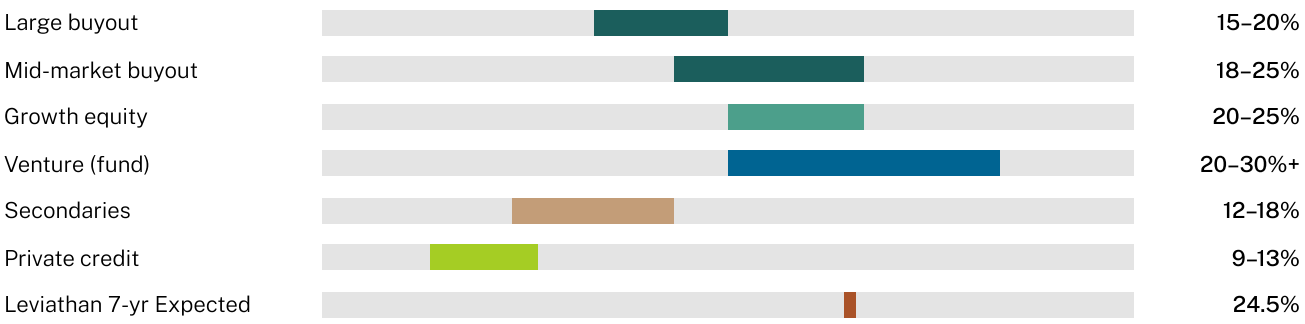
13 Formula Cheat Sheet

Item	Formula
Ownership	Check ÷ Post-money
Post-money	Pre-money + New money
Single-exit IRR	MOIC^(1/T) − 1
Rule of 72	Years to double ≈ 72 ÷ IRR(%) (2x in 5 years ≈ 15%; 3x in 5 years ≈ 24.6%; 3x in 7 years ≈ 17%)
TVPI	DPI + RVPI
KS-PME (single cash flow)	MOIC ÷ (1 + index return)^T
Compounded pref	Capital × [(1 + h)^T − 1]
Full catch-up amount	Pref × carry ÷ (1 − carry)
Equity value	EV − Net debt − Preferred − Minorities − Debt-like items
LBO equity at exit	Exit EBITDA × Exit multiple − Exit net debt
Value bridge	ΔEBITDA × Entry multiple + ΔMultiple × Exit EBITDA + Debt paydown
Retention	Π (1 − dilution per round)
Probability-weighted value	$\Sigma\ p_i \times Value_i$
▸ Leviathan proceeds	Check × (1 + CAGR)^Hold × Retention
▸ Shock CAGR	Base CAGR + (Composite − 2.5) × 8%
▸ Munitions-adjusted CAGR	CAGR + avg(Shortage, FMS) × uplift per point
▸ Pre-IPO entry	Forward EBIT × IPO multiple × (1 − pre-IPO discount)

14 Quick-Reference Return Targets (rule-of-thumb, net to LP)[4]

Target net IRR ranges[4]

axis 5% to 35%; bar spans the rule-of-thumb range



The Leviathan marker is the probability-weighted 7-year holdco net IRR from the IC model.[1]

Strategy	Target net IRR	Target net MOIC	Hold
Large buyout	15–20%	1.8–2.2x	4–6 years
Mid-market buyout	18–25%	2.0–2.5x	4–6 years
Growth equity	20–25%	2.5–3.0x	4–7 years
Venture (fund level)	20–30%+	3.0x+ (top quartile)	8–12 years
Secondaries	12–18%	1.5–1.8x	3–5 years
Private credit	9–13%	1.3–1.5x	3–5 years
▸ Leviathan holdco (7-year, Expected)	**24.5%**	**4.63x**	7 years

Notes

Sources and endnotes

1. Leviathan core PE holdco model, January 2027 internal version ($1,406M). All ▸ figures (holdco size, sleeve structure, net IRR / MOIC, KS-PME, retention, entry marks) are model outputs, estimated and unaudited.
2. Worked examples (waterfall, LBO, EV bridge, dilution) use round illustrative numbers chosen for arithmetic clarity, not market data.
3. Capital-stack pricing is indicative and varies with the rate environment, leverage and credit quality.
4. Return targets are rule-of-thumb ranges used in industry practice, not sourced performance data.
5. Regulatory thresholds and rules are summarized for reference; confirm against current statute and counsel before relying on them.

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Note: Model figures are estimated and unaudited. See Sources and endnotes 1–5.